

Azmani Sultana

Id: 22201949

Section 03

Ann no 1

Greedy Best First Search:

A^{10}

$A^{10} \rightarrow B^8 F^6$

$F^6 \rightarrow B^8 G^5 H^3$

$H^3 \rightarrow B^8 G^5 I^1$

$I^1 \rightarrow B^8 G^5 E^3 J^0$

$J^0 \rightarrow B^8 G^5 E^3$

$A^{10} \rightarrow F^6 \rightarrow H^3 \rightarrow I^1 \rightarrow J^0$

$$\text{cost} = 3 + 7 + 2 + 3 = 15$$

A* search

$$F(A) = 0 + 10 = 10$$

$$F(B) = 6 + 8 = 14$$

$$F(F) = 3 + 6 = 9$$

$$F(G) = (3+1) + 5 = 9$$

$$F(H) = (3+7) + 3 = 13$$

$$F(I) = (3+1+1) + 3 = 8$$

$$F(J) = (3+1+3) + 1 = 8$$

$$F(E) = (3+1+3+5) = 12$$

$$F(D) = (3+1+3+3) = 10$$

$$F(C) = (3+1+1+2) + 1 = 7$$

$$F(B) = (3+1+1+2+5) = 12$$

$$F(A) = (3+1+1+2+3) = 10$$

$A^{10} \rightarrow F^9 \rightarrow G^9 \rightarrow H^8 \rightarrow I^8 \rightarrow J^{10}$

$$\therefore \text{cost} = 3 + 1 + 1 + 2 + 3 = 10$$

A^{10}

$A^{10} \rightarrow B^{14} F^9$

$F^9 \rightarrow B^{14} G^9 H^{13}$

$G^9 \rightarrow B^{14} H^{13} H^8 I^8$

$I^8 \rightarrow B^{14} H^{13} H^8 E^{15} J^{10}$

$H^8 \rightarrow B^{14} H^{13} E^{15} J^{10} I^8$

$I^8 \rightarrow B^{14} H^{13} E^{15} J^{10} E^{15} J^{10}$

$J^{10} \rightarrow B^{14} H^{13} E^{15} J^{10} E^{15}$

Ans no 2

a) Greedy Best First Search:

A^{20}

$A^{20} \boxed{B^{16} D^{10}}$

$D^{10} \boxed{B^{16} C^6 E^4}$

$E^4 \boxed{B^{16} C^6 G^0}$

$G^0 \boxed{B^{16} C^6}$

$A^{20} \rightarrow D^{10} \rightarrow E^4 \rightarrow G^0$

$$\text{cost} = 6 + 10 + 6 = 22$$

A* search

$$F(A) = 0 + 20 = 20$$

$$F(B) = 2 + 16 = 18$$

$$F(D) = 6 + 10 = 16$$

$$F(C) = (6 + 7) + 6 = 19$$

$$F(E) = (6 + 10) + 4 = 20$$

$$F(C) = (2 + 5) + 6 = 13$$

$$F(D) = (2 + 5 + 7) + 10 = 24$$

$$F(E) = (2 + 5 + 6) + 4 = 17$$

$$F(G) = (2 + 5 + 9) + 0 = 16$$

$A^{20} \rightarrow B^{18} \rightarrow C^{13} \rightarrow G^{16}$

$$\text{cost} = 2 + 5 + 9 = 16$$

A^{20}

$A^{20} \boxed{B^{18} D^{16}}$

$D^{16} \boxed{B^{18} C^{19} E^{20}}$

$B^{18} \boxed{C^{19} E^{20} C^{13}}$

$C^{13} \boxed{C^{19} E^{20} D^{24} E^{17} G^{16}}$

$G^{16} \boxed{C^{19} E^{20} D^{24} E^{17}}$

b)

Admissibility:

$$h(D) \leq h^*(D)$$

$$10 \leq 7 + 9 = 16 \quad (D \rightarrow C \rightarrow G)$$

$$10 \leq 10 + 6 = 16 \quad (D \rightarrow E \rightarrow G)$$

So, current heuristic value of D is admissible.

consistency:

$$h(D) \leq c(D, n') + h(n')$$

$$h(D) \leq c(D, A) + h(A)$$

$$10 \leq 6 + 20$$

$$\Rightarrow 10 \leq 26$$

$$h(D) \leq c(D, C) + h(C)$$

$$10 \leq 7 + 6$$

$$\Rightarrow 10 \leq 13$$

$$h(D) \leq c(D, E) + h(E)$$

$$10 \leq 10 + 4$$

$$\Rightarrow 10 \leq 14$$

So, $h(D)$ is consistent.

The current heuristic value of D is both consistent and admissible.

Ann no 3

A* search (tree search)

$$F(S) = 0 + 7 = 7$$

$$F(A) = 3 + 9 = 12$$

$$F(D) = 2 + 5 = 7$$

$$F(B) = (2+1) + 4 = 7$$

$$F(E) = (2+4) + 3 = 9$$

$$F(C) = (2+1+2) + 2 = 7$$

$$F(F) = (2+1+1) + 3 = 7$$

$$F(G) = (2+1+2+4) + 0 = 9$$

$$F(H) = (2+1+1+3) + 0 = 7$$

S^7

S^7 $A^{12} D^7$

D^7 $A^{12} B^7 E^9$

B^7 $A^{12} E^9 C^7 E^7$

C^7 $A^{12} E^9 E^7 G^9$

E^7 $A^{12} E^9 G^9 G^7$

G^7 $A^{12} E^9 G^9$

$$S^7 \rightarrow D^7 \rightarrow B^7 \rightarrow E^7 \rightarrow G^7$$

$$\text{cost} = 2 + 1 + 1 + 3 = 7$$

Greedy Best First search

S^7

S^7 $A^9 D^5$

D^5 $A^9 B^4 E^3$

E^3 $A^9 B^4 G^0$

G^0 $A^9 B^4$

$$S^7 \rightarrow D^5 \rightarrow E^3 \rightarrow G^0$$

$$\text{cost} = 2 + 4 + 3 = 9$$